



THE SCIENCE

Formulas, Notation & Logic

Every number the system reports, derived. We measure the way the 2025 conversation-intelligence and LLM-evaluation literature does — faithfulness and speaker separation — not a brittle exact-match "accuracy".

1 • Diarization — channels, not guessing

Let **L** = agent channel, **R** = customer channel. We transcribe each separately into turn sets **T_L** and **T_R**, then rebuild the real conversation:

```
T = sort( T_L ∪ T_R )    ordered by turn.start
```

Separation quality

To prove the two channels really are separate speakers, take the Pearson correlation **ρ** between the two channels' energy envelopes **e_L**, **e_R** (100 ms windows):

$$\rho = \frac{\sum (e_{L_i} - \bar{e}_L)(e_{R_i} - \bar{e}_R)}{\sqrt{[\sum (e_{L_i} - \bar{e}_L)^2 \cdot \sum (e_{R_i} - \bar{e}_R)^2]}}$$

$$\text{separation_quality} = 1 - \max(0, \rho)$$

Low (even negative) **ρ** ⇒ the speakers don't overlap ⇒ attribution is correct by construction. Measured on this dataset: **ρ ≈ -0.13** → **separation_quality = 1.00** (100%) .

→ Formulation & Notations:-

Diarization - mly channels, not guessing

let

'L' → agent

'R' - Customer

we transcribe each separately, giving turn sets like 'T_L' and 'T_R'

so transcript the final one is like

$T = \text{sort}(T_L \cup T_R)$ order by turn start

→ separation quality :-

let me prove the channels really are separate speaker by the person correlation

'p' between the two 'channels' energy envelopes "e_L", "e_R" (looks window):-

so formulation:-

$$\rho = \frac{\sum (e_{-L_i} - \bar{e}_{-L})(e_{-R_i} - \bar{e}_{-R})}{\sqrt{[\sum (e_{-L_i} - \bar{e}_{-L})^2 \cdot \sum (e_{-R_i} - \bar{e}_{-R})^2]}}$$

$$\text{separation_quality} = 1 - \text{Max}(0, \rho)$$

Low (even negative) ' ρ ' $\Rightarrow$ the two speakers
don't overlap $\Rightarrow$ attribution is correct &&
by construction"

measured on this dataset that given

$$\rho \approx -0.13$$

$$\text{separation_quality} = 1.00 \text{ (100\%)} \text{ nearly.}$$

$\rightarrow$ Evidence verification -

normalise $c(s)$ = $\text{count}(c(s))$ with only [a-zo-g]

verified $(e_j) = 1$ if $\text{normalise}(q_{\text{all}}) \leq \text{normalise}$
(z-test) or $|Q_{HB}| / |Q| \geq 0.60$

o otherwise

where Q = words (quote), B = words (z-test)

2 • Evidence verification & faithfulness

Each judgment j carries evidence $e_j = (\text{timestamp}, \text{quote}, \text{speaker})$. Verification finds the transcript turn τ at that timestamp+speaker, normalises both, and accepts if the quote appears in the turn *or* shares $\geq 60\%$ of its significant words:

```
normalise(s) = lowercase(s) keeping only [a-z0-9 ]

verified(e_j) = 1  if  normalise(quote)  $\subseteq$  normalise( $\tau$ .text)
                or  |Q  $\cap$  B| / |Q|  $\geq$  0.60
                0  otherwise      (Q = words(quote), B = words( $\tau$ .text))
```

Per-call evidence score (n = judgments checked):

```
evidence_score = (1/n)  $\cdot$   $\sum_j$  verified(e_j)
```

System-wide faithfulness (over all calls C):

$$\text{faithfulness} = \frac{\sum_c \sum_j \text{verified}(e_{\{c,j\}})}{\sum_c (\# \text{ judgments in } c)} = 0.999 \quad (99.9\%)$$

per call evidence score.

$$\text{evidence score} = \frac{1}{n} \times \sum_j \text{verified}(e-j)$$

system wide faithfulness (over all calls c);

$$\text{faith} = \frac{\sum_c \sum_j \text{verified}(e-\{c,j\})}{\sum_c (\# \text{ judgments inc})}$$

$$= 0.999 \quad (99.9\%)$$

Attention Score (Need)

$$A \in [0, 100]$$

Hand-worked: per-call evidence score, system-wide faithfulness, and the attention score.

3 • Needs-attention score $A \in [0, 100]$

Additive signals, then clamped. `neg` = # negative cues from the customer.

```
A = 10                                (base)
  + min(35, 12·neg)                    (customer negativity)
  + 30 escalated / 28 unresolved / 15 unclear
  + 20 if mood shifted to negative (-5 if recovered to positive)
  + 22 if HIGH_RISK topic / 10 if MED_RISK
  + 12 if agent repeated a question
  + 10 if dur ≥ 90s (5 if ≥ 70s)
A = clamp(A, 0, 100)

HIGH_RISK = {fraud_dispute, card_lost_stolen, complaint, account_access}
MED_RISK  = {transaction_issue, fees_charges, loan_mortgage, payment_transfer}
```

4 • QA / compliance score $Q \in [0, 100]$

Each check `k` has a pass flag `p_k ∈ {0,1}` and a weight `w_k`:

CHECK	WEIGHT
Greeting (opened professionally)	1
Identity verified before a sensitive action	3 (0 if not required)
Action confirmed	2
Empathy shown	1
No repeated question	1
Proper close	1

$$Q = \frac{\sum_k p_k \cdot w_k}{\sum_k w_k} \times 100 \quad (\text{over checks with } w_k > 0)$$

```
resolution_risk =
  (sensitive_action ∧ ¬identity_verified)           # compliance risk
  ∨ (sounded_resolved ∧ customer_asked ∧ ¬action_confirmed ∧ status ≠ resolved)
```

QA / Compliance score $Q \in [0, 100]$

$$Q = \frac{\sum_k p_k \cdot w_k}{\sum_k w_k} \cdot 100$$

Over checks with $w_k > 0$

Number of quate (full run 1441 calls)

Parameter	Score
calls	1441
customers	100
agents	10
fairness	99.9%
disruption	100%
convergence	100%
resolution risk	195

Hand-worked: the QA score, and the measured results on all 1,441 calls.

Measured results (full 1,441-call run)

PARAMETER	VALUE
Calls	1,441
Customers	100
Agents	10
Faithfulness	99.9%
Diarization separation	100%
Coverage	100%
Resolution-risk calls	195

Companion for CallRadar — Formulas & Notation. See also [Architecture](#), the complete documentation. the complete documentation.nbsp;|the complete documentation.nbsp; Team: [Sahil Koshriya](#) · [Sakshi](#).